FORMAL METHOD

Assignment 1

Total Marks: 50 Obtained Marks:	Date:
SECTION:	Name 1: Name 2:

CLO #	CLO	Taxonomy Level	Questions Mapped	Total Marks	Marks Obtained
CLO 1	Understand the problem-solving skills using mathematical logics such as propositional logic, predicate logic and temporal logics.	C2	1	26	
			2	14	
			3	10	

Late Submission Policy:

- A penality of 25% per day will be applied to late submission.

Instructions:

- The use of AI tools in completeing the assignments is strictly prohibited. Any submission will result in zero marks.

Question 1:

- If Set A = {1, 2, 3, 4, 5} and Set B = {4, 5, 6, 7}, find the Union and Intersection.
- What is the Set Difference (A minus B) if A = {red, blue, green} and B = {green, yellow}?
- Find the Symmetric Difference of Set X = {10, 20, 30} and Set Y = {30, 40, 50}.
- If the Universal Set is {1 to 10} and Set A = {2, 4, 6, 8}, what is the Complement of Set A?
- Write all the elements of the Power Set of S = {apple, banana}.
- If Set A is a Subset of Set B, what is the result of A Intersection B?
- What does the Idempotent Law state for the Union of a set?
- How do you express "A minus B" using Intersection and Complement?

Question 2:

A tech firm is reviewing the skills of its 120 Software Engineers. They are testing for proficiency in three modern domains: Java (J), Python (P), and JavaScript (JS).

- Total Engineers (U) = **120**
- Proficient in Java (J) = **70**
- Proficient in Python (P) = **60**
- Proficient in JavaScript (JS) = **50**
- Proficient in Java and Python ($J \cap P$) = **30**
- Proficient in Python and JavaScript ($P \cap JS$) = **25**
- Proficient in Java and JavaScript ($J \cap JS$) = **20**
- Proficient in **All Three** ($J \cap P \cap JS$) = **10**

1. Calculate the number of engineers who are proficient in?
2. Find the number of engineers who know exactly two languages.?
3. Calculate the total number of engineers who know at least one of these three languages (The Union).
4. How many engineers know none of these languages and require immediate training?

Question 3:

- You are designing a software system for an Automated Teller Machine (ATM). Define what "Formal Specification" means in this context and list four desirable properties that your specification must satisfy to ensure the system is "mathematically sound."